

Forecasting & Evaluation — Detailed Guide

Retail Demand Forecasting (Walmart M5, Hierarchical) · Team Study Material

1. Where this topic starts and stops

The ML Engineer answers: *"How did we build and train the prediction model?"*

This topic answers something different:

"How do we use the trained model to produce a 28-day forecast, how should that forecast be read, and how do we prove the numbers are trustworthy?"

[ML ENGINEER]		[THIS TOPIC]
train the models	->	produce the forecast
		interpret the forecast
		measure how wrong it is
		prove it reproduces

If a judge asks "what is Tweedie", that is the ML Engineer. If a judge asks "how accurate is this, really?" — that is you.

What we forecast

Verified: units sold, per product, per store, per day, for **28 days ahead** — d_1942 ... d_1969 (2016-05-23 → 2016-06-19), for all **30,490** store-item series.

That is $30,490 \times 28 = 853,720$ numbers in the delivered forecast.

Horizon. How far ahead we predict. Ours is 28 days, set by `HORIZON = 28` in `03_RESEARCH/pipeline/config.py` and by the competition's submission format (columns `F1...F28`). **Forecast origin.** The last day whose sales we are allowed to know. Ours is `d_1941 = 2016-05-22`.

2. Direct vs recursive forecasting

There are two ways to get 28 days out of a model, and we use **both**.

Direct

One model predicts **all 28 days at once**. horizon (1...28) is a feature, so the model knows whether it is being asked about tomorrow or four weeks out.

```
origin d_1941 ■■■> day 1
               ■■> day 2
               ■■> ...
               ■■> day 28    all from the same frozen inputs
```

- **Good:** no error accumulation — day 28 is predicted directly, not through 27 intermediate guesses.
- **Bad:** every day sees the same frozen history, so it cannot react to what "happened" on day 1.

Recursive

One model predicts **one day**. That prediction is written back into the history as if it were real, and the model predicts the next day. Repeat 28 times.

```
origin ■> day 1 ■■
               ■■> feed back ■> day 2 ■■
                           ■■> feed back ■> day 3 ... 28
```

- **Good:** each step uses an updated picture, so momentum carries forward.
- **Bad: error accumulation** — a wrong day 1 corrupts every later day.

This is exactly why the recursive member drops the recency features. A prediction of 0.3 fed back as "a sale happened" would reset days_since_last_sale and corrupt the rollout.

Side by side

	Direct	Recursive
Predicts	all 28 days at once	1 day, 28 times
Uses horizon feature	yes	no
Features	38	32
Error accumulation	none	yes
Verified RMSE alone	2.1211	2.1185

3. The blend (ensemble)

Ensemble. Combining several models' predictions into one. It works when the models make **different** mistakes — the errors partly cancel.

Our verified blend:

```
FINAL = clip( 0.60 x DIRECT + 0.40 x RECURSIVE , 0, none )
```

`clip(..., 0)` means negatives become zero — you cannot sell -0.3 units.

Model	RMSE	MAE
Direct alone	2.1211	1.0293
Recursive alone	2.1185	1.0689
Blend 0.60/0.40	2.0929	1.0395

The blend beats **both** members. That only happens because they are architecturally different — their **residual correlation is 0.9488**. High, but not 1.0, and that gap is where the improvement comes from.

How 0.60 was chosen — and why this matters. The weight was optimised on an **inner window** (origin `d_1885`, targets `d_1886-d_1913`) which ends **before** the evaluation window begins. It was never fitted on the data used to report accuracy. Fitting a blend weight on your evaluation window is a classic way to report a score you cannot reproduce.

4. Hierarchical forecasting

Hierarchy. Products roll up: an item in a store belongs to a department, a category, a store, a state, and the whole chain.

We forecast at the **bottom level** (store-item-day) and **sum upward**. Because aggregates are exact sums of the bottom-level forecasts, they are automatically **coherent** — the store total always equals the sum of its products. Nothing can disagree with itself.

Verified accuracy at all 12 levels (1 – WAPE, primary window):

Level	Groups	RMSE	Accuracy
L1 total (whole chain)	1	1589.4663	97.2%
L2 state	3	741.6479	96.0%
L4 category	3	848.6328	96.2%
L5 department	7	474.5415	94.8%
L6 state-category	9	379.9640	94.9%
L3 store	10	305.9148	94.7%
L7 state-department	21	210.5520	93.2%
L8 store-category	30	154.6778	93.1%
L9 store-department	70	87.6021	91.1%
L10 item (all stores)	3,049	8.5652	70.8%
L11 item-state	9,147	4.2026	55.5%
L12 store-item	30,490	2.0929	27.9%

This table is the single most important thing in this topic.

Why does accuracy improve so dramatically going up? Because errors are largely independent and **cancel** when summed. If one store over-predicts by 2 and another under-predicts by 2, the chain total is exactly right. At the bottom level there is nothing to cancel against — and 54% of individual store-item-days are zero, so a single-unit miss is a huge relative error.

The honest framing: 27.9% at the bottom is not a bad model. It is what per-product-per-day accuracy genuinely looks like on intermittent retail demand. **Always quote the level that matches the decision being made** — a warehouse planner cares about L3/L5, a shelf replenisher cares about L12.

5. The metrics, explained properly

RMSE — Root Mean Squared Error

```
RMSE = sqrt( mean( (actual - predicted)^2 ) )
```

- actual = what really sold
- predicted = what we said would sell
- square the difference, average it, take the square root

Squaring means **big mistakes count much more than small ones**. Being wrong by 10 is 100 times worse than being wrong by 1, not 10 times. RMSE is in the same units as sales.

VERIFIED: RMSE = 2.0929 on 853,720 predictions.

MAE — Mean Absolute Error

```
MAE = mean( |actual - predicted| )
```

The average size of the mistake, ignoring direction. No squaring, so a few huge misses do not dominate.

VERIFIED: MAE = 1.0395 — the typical miss is about one unit per product per day.

***RMSE vs MAE — know this comparison.** | | RMSE | MAE | |---|---|---| | Punishes large errors | heavily | proportionally | | Sensitive to outliers | yes | no | | Use when | big misses are costly | you want the typical miss | RMSE 2.09 is bigger than MAE 1.04 — that gap tells you a minority of large errors is doing a lot of the damage. On spiky retail data that is expected.*

WAPE — Weighted Absolute Percentage Error

```
WAPE = sum(|actual - predicted|) / sum(actual)
```

Total error divided by total volume — "our mistakes add up to X% of everything sold". We use WAPE rather than MAPE because MAPE divides by the actual value, and **68% of our actuals are zero** — dividing by zero is undefined. WAPE never has that problem.

VERIFIED: WAPE = 0.7205 at store-item level → accuracy = 1 – 0.7205 = **27.9%**.

Bias

```
Bias = mean(predicted - actual)
```

Are we systematically over- or under-predicting? Zero is ideal.

VERIFIED: Bias = -0.0224 — we under-predict by about 0.02 units per prediction on average. That is very close to neutral, which matters for inventory: a persistent bias would silently accumulate into a stock problem.

The occurrence metrics — a different question

RMSE and MAE ask "how many units?". There is a second, simpler question: **"will this product sell at all today?"** That is a yes/no question, so it needs classification metrics.

***Our rule (verified):** actual event = actual sales > 0. Predicted event = point forecast ≥ 0.5 units (i.e. it rounds to at least one unit).*

Imagine a 2x2 table of predicted vs actual:

	actually sold	actually didn't
we said it would	true positive	false positive
we said it wouldn't	false negative	true negative

- **Accuracy** = how often we were right overall
- **Precision** = of the times we said "it will sell", how often it did
- **Recall** = of the times it actually sold, how often we caught it
- **F1** = the harmonic mean of precision and recall — one number balancing both

VERIFIED values on 853,720 predictions (threshold 0.5):

Metric	Value
Accuracy	0.6980
Precision	0.6321
Recall	0.8068
F1	0.7088
Base rate (how often anything sells)	0.4556

How to read this. Recall (0.8068) is much higher than precision (0.6321): we catch most real sales, but we also flag a lot of days that turn out to be zero. For retail that is usually the safer direction — missing a sale costs a customer, predicting a sale that does not happen costs a little shelf space.

WARNING — these four are very easy to transpose. Some earlier project briefs quoted accuracy as 0.8068 and precision as 0.7088. Those are the **recall** and **F1** values. The numbers in the table above are the ones computed from the backtest artefact and served by GET /api/v1/accuracy/occurrence. **Use these.**

6. How error varies — across days and across products

The forecast is not equally good on every day.

Horizon day	RMSE	MAE
Day 1	1.8106	0.9038
Day 2	1.6499	0.8716
Day 3	1.7272	0.8846
...		
Day 26	1.9520	1.0237
Day 27	2.2990	1.1739
Day 28	2.3807	1.2053

Error is roughly flat early and rises toward the end of the horizon — day 28 is about 31% worse than day 1 on RMSE. That is exactly what you would expect and it is worth saying out loud: *"a forecast four weeks out is genuinely less reliable than tomorrow's, and we can show you by how much."*

Different products are different problems

We classify every series by its demand pattern (Syntetos-Boylan regimes).

Regime	Series	RMSE
Intermittent	23,246	1.2518
Lumpy	4,276	2.9419
Smooth	2,271	4.1029
Erratic	694	5.6164
Never sold	3	0.1570

Note the counts: **23,246 of 30,490 series are intermittent** — they sell sporadically. The "smooth" and "erratic" series have *higher* RMSE simply because they sell more units, so there is more to be wrong about.

7. Forecast ranges — and what they are not

The product shows a **planning range** around each forecast. Be very careful how you describe it.

*It is **NOT** a confidence interval or a prediction interval. Our model emits a single point forecast; it does not produce a probability distribution.*

What the range actually is: **observed past error**. We took the residuals (actual – predicted) from 8 held-out backtest windows, normalised them by $\sqrt{\max(\text{forecast}, 1)}$ to account for larger products having larger errors, grouped them by demand regime and horizon day, and read off empirical percentiles.

VERIFIED: measured coverage is 90.0% — the band contained the actual value 90% of the time in backtesting.

Every API response carries a `band_basis` field stating in words that this is observed error, not a model-produced interval. That wording is deliberate and tested.

8. Validation methodology

Fixed-origin, direct 28-day backtest. Stand at origin T^* , freeze every history-derived feature, predict $T+1 \dots T+28$, compare against what actually happened.

Block	Days	Dates
Training	d_1 ... d_1913	2011-01-29 ... 2016-04-24
Validation	d_1914 ... d_1941	2016-04-25 ... 2016-05-22
Final forecast	d_1942 ... d_1969	2016-05-23 ... 2016-06-19

Multiple windows, not one. A single window can flatter you. Verified results at the shipped blend weight:

Window	Dates	RMSE	MAE
primary_spring_2016	2016-04-25 → 05-22	2.0929	1.0395
christmas_2015	2015-12-12 → 2016-01-08	2.1547	0.9446
summer_2015	2015-07-16 → 08-12	2.1074	0.9909
autumn_2015	2015-10-02 → 10-29	2.1496	1.0126

The blend beat the direct member on **all four**, mean $\Delta\text{RMSE} = -0.0242$. The API exposes **8** backtest windows in total via `GET /api/v1/accuracy/windows`.

Baselines on the same window: naive "repeat the last 28-day average" scores RMSE 2.2430; predicting zero everywhere scores 3.9161. Always quote a baseline — a number alone means nothing.

The rule we never break

No accuracy figure is ever quoted for the delivered forecast window.

d_1942 ... d_1969 has **no recorded ground truth** in any file. Nobody knows what actually sold. Every accuracy number in this guide and in the product comes from **held-out validation windows** where the true outcome is known.

The frontend enforces this: the Forecast page shows no accuracy, and the Validation page — which does — is explicitly about historical windows. It is a small discipline that judges notice.

9. Frozen forecast verification

Two independent guarantees.

1. Live re-inference. `POST /api/v1/inference/verify` reloads both frozen LightGBM models, rebuilds the features, re-runs the blend at 0.60/0.40, and compares against the shipped forecast file.

VERIFIED RESULT: `max_abs_diff = 0.000e+00` across all 853,720 predictions. Not "close". Identical.

2. Regression tests. `01_PROJECT/backend/tests/test_integrity.py` checks that model file hashes match the manifest, that served forecasts equal the frozen CSV row-for-row, that the chain total matches, that the cached backtest still reproduces **RMSE 2.0929 / MAE 1.0395**, that `p_blend` still equals `0.60·direct + 0.40·recursive`, and that the error bands still cover ~90%.

Plus: the leakage suite passes **46/46**, and during the Use Case 11 audit all four windows were **retrained from scratch** and reproduced to four decimals.

10. Common misunderstandings

"27.9% accuracy means the model is bad." No. That is per-product-per-day on data where 54% of cells are zero. The same model is 97.2% accurate chain-wide. Quote the level that matches the decision.

"The shaded band is a confidence interval." No. It is observed past error, measured at 90.0% coverage.

"You can tell us how accurate next month's forecast will be." No. That window has no ground truth. We can tell you how accurate the model *was* on held-out windows.

"A forecast of 0.6 units is wrong." It is an expected value — about 17 units over 28 days.

"RMSE and MAE measure the same thing." No. RMSE punishes large errors much harder. The gap between 2.09 and 1.04 is itself informative.

"Higher recall is always better." Recall 0.8068 with precision 0.6321 is a trade-off we accept, not an accident.

11. How to present this

"My part is what happens after the model exists — producing the forecast and proving it's trustworthy.

We forecast 28 days ahead for 30,490 product-store combinations, which is 853,720 numbers. We do it two ways and combine them: a direct model that predicts all 28 days at once, and a recursive one that predicts a day and feeds it back in. Alone they get RMSE 2.1211 and 2.1185; blended 60/40 we get 2.0929 — better than either, because they're different architectures making different mistakes.

On metrics: RMSE is 2.0929, which punishes big misses because errors are squared. MAE is 1.0395 — the typical miss is about one unit per product per day. Bias is -0.0224 , so we're not systematically over or under.

The number I'd want you to see is the hierarchy table. At the individual product-store-day level we're 27.9% accurate. At store level 94.7%, and chain-wide 97.2%. That's not us hiding a bad number — it's that errors cancel when you aggregate, and 54% of individual cells are zero. We always show the level that matches the decision you're making.

And we never quote accuracy for the delivered forecast, because that window has no ground truth yet. Everything we report is from held-out windows.

Finally, we can prove the model still produces exactly what we shipped — an endpoint reloads it, re-runs all 853,720 predictions, and the maximum absolute difference is exactly zero."

12. Likely judge questions

Beginner

Q: What does RMSE 2.09 mean? A: On average, weighting big mistakes more heavily, we're off by about 2 units per product per day. For comparison, guessing zero everywhere gives 3.92.

Q: Is 27.9% accuracy good? A: For per-product-per-day on intermittent demand, yes. The same forecast is 94.7% accurate at store level and 97.2% chain-wide. It depends entirely on the level you're deciding at.

Q: How far ahead do you forecast? A: 28 days — 2016-05-23 to 2016-06-19, for all 30,490 series.

Technical

Q: Why WAPE instead of MAPE? A: MAPE divides by the actual value, and 68% of our actuals are zero — undefined. WAPE divides total error by total volume, so it's always defined.

Q: How did you choose the 0.60 blend weight? A: Optimised on an inner window ending at `d_1913`, strictly before the evaluation window. It was never fitted on the data we report accuracy on.

Q: Why does the blend beat both members? A: Their residual correlation is 0.9488 — high, but not 1. They're architecturally different, so some error cancels.

Q: What is the shaded band on the chart? A: Observed past error from 8 backtest windows, normalised by $\sqrt{\text{forecast}}$ and grouped by regime and horizon day. Measured coverage 90.0%. It is explicitly not a prediction interval — our model emits point forecasts only.

Q: How do we know the forecast wasn't just hand-tuned? A: Live re-inference reloads the frozen models and reproduces the shipped forecast with `max_abs_diff = 0.000e+00`, and a regression test re-checks RMSE 2.0929 / MAE 1.0395 on the cached backtest.

Q: What's your accuracy on next month? A: Unknowable — that window has no ground truth. Our best honest estimate is the held-out validation result: RMSE 2.0929.

Q: Does error grow over the horizon? A: Yes — day 1 RMSE is 1.8106, day 28 is 2.3807, about 31% worse. We publish it rather than averaging it away.

13. Glossary

Term	Meaning
Horizon	How many days ahead; ours is 28
Forecast origin	Last day whose sales we may know (d_1941)
Direct forecasting	Predict all 28 days at once
Recursive forecasting	Predict one day, feed it back, repeat
Ensemble / blend	Combining models; ours is 0.60 direct + 0.40 recursive
Residual	actual – predicted
Coherent	Aggregates are exact sums of their parts
RMSE	Root Mean Squared Error — punishes big misses
MAE	Mean Absolute Error — the typical miss
WAPE	Total error ÷ total volume
Bias	Systematic over- or under-prediction
Precision	Of predicted sales, how many were real
Recall	Of real sales, how many we caught
F1	Harmonic mean of precision and recall
Base rate	How often the event happens at all (0.4556)
Backtest	Testing on historical data with known outcomes
Ground truth	The actual recorded outcome